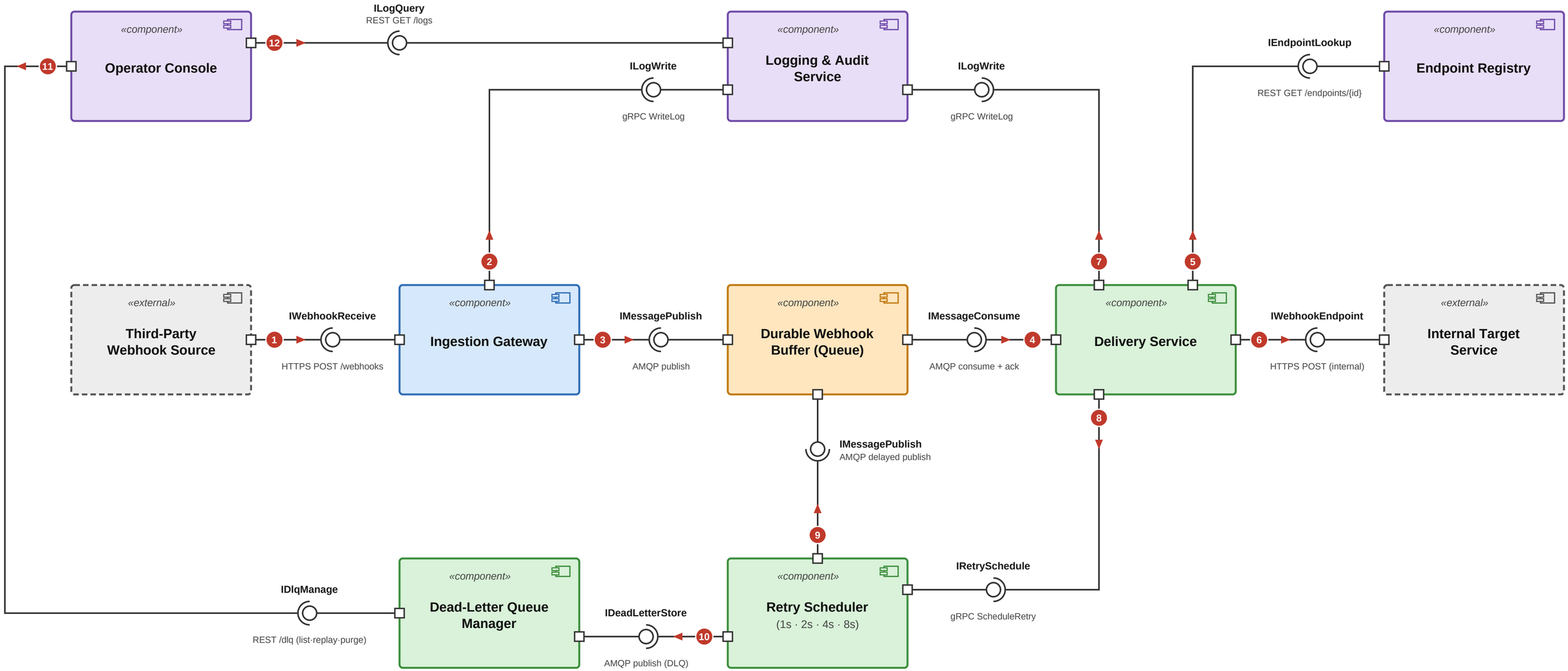
Webhook Ingestion & Retry Mechanism Hub — UML Component Diagram

Microservices architecture · Lab 3: Component Modelling & Architectural Pattern Selection · Sanjana Aravamudhan (PES1UG24CS713)



Legend

- Provided interface (ball) — service offered
- Required interface (socket) — service needed
- Port (interaction point on component boundary)
- n — Step n of the webhook flow, arrow = direction of data flow

- Ingestion (public-facing)
- Messaging / buffering
- Delivery & reliability
- Platform & operations
- External system (dashed)

Data flow & interactions

1 Third-Party Source → Gateway: signed webhook payload (HTTPS POST)

2 Gateway → Logging: received request payload (FR-002)

3 Gateway → Buffer: payload enqueued durably, then 202 Accepted (FR-001, NFR-001)

4 Buffer → Delivery: queued webhook pulled for delivery

5 Delivery → Registry: configured target endpoint looked up (FR-003)

6 Delivery → Target: payload forwarded to internal service (FR-003)

7 Delivery → Logging: delivery success / failure and retry activity

8 Delivery → Retry Scheduler: failed delivery handed over (FR-004)

9 Retry Scheduler → Buffer: re-enqueued after 1s, 2s, 4s, 8s delay (FR-004)

10 Retry Scheduler → DLQ Manager: persisted once retries are exhausted (FR-005)

11 Operator → DLQ Manager: inspect, replay or purge dead letters

12 Operator → Logging: query delivery and retry activity